

USAFA's mandate includes providing an exceptional education and recognition as an elite academic institution. The Engineering Division maintains ABET accreditation as a credible validation of USAFA's educational excellence. The Permanent Professor of Astronautics ensures accreditation by hiring qualified faculty and ensuring the Department of Astronautics satisfies ABET guidelines. My education and experience qualify me to teach Astronautical Engineering and have uniquely prepared me to lead the Department in ongoing excellence.

I have three relevant ABET-accredited engineering degrees.

- BS in Engineering Mechanics (USAFA, 2006) provided a foundation for engineering thinking and prepared me for hands-on engineering in Air Force assignments.
- MS in Astronautical Engineering (AFIT, 2011) introduced me to the basics of space engineering.
- PhD in Aerospace Engineering (NCSU, 2019) gave me the freedom to pursue independent learning and deep expertise in spacecraft component testing.

I have engineering education experience from two teaching assignments at USAFA (2014–2016, 2022–present), including instruction, assessment, and curriculum development.

- I instructed six courses in two engineering departments and served as course director for four of them.
- As assessment officer, I created a mid-cycle Department review and annual assessment reports to support the next ABET visit.
- I led the Engineering Division's 2024 Engineering Program Advisory Council (EPAC), an external review to ensure USAFA's engineering programs meet AF and SF needs.

My non-academic jobs have also prepared me to credibly lead an ABET-accredited engineering program.

- At Holloman AFB, I learned engineering practice performing hands-on work in flight test and rocket sled test.
- At Los Angeles AFB, I learned space acquisition and operations and cemented space lessons from AFIT.
- In a DIA assignment in Charlottesville VA, I gained invaluable leadership experience, reconstituting a 29-person branch after COVID while addressing long-standing cultural problems.

ABET accreditation provides more than just credibility; it actively improves the major and leads to better-prepared graduates. The accreditation cycle itself provides a framework for assessing and improving curriculum. In addition, evaluators provide expert feedback on

program strengths and weaknesses. Finally, it provides engineering graduates with an ABET-accredited degree—a prerequisite for admission to nearly all graduate engineering programs in America.

Sustaining the culture of excellence necessary for accreditation requires continuous work. First, the department must continually recruit and develop qualified faculty members. In addition, the department must embrace engineering advances while maintaining excellence in fundamentals. Military assignment cycles provide a constant infusion of fresh engineering practices, but incorporating advances takes sustained and focused effort. I have significant experience identifying improvements and updating coursework to keep pace with engineering practice.

As one example, while serving as Division EPAC coordinator I identified communication as a curriculum shortfall across multiple programs. To address this shortfall, I pursued a three-semester project to incorporate new material into the Astronautical Engineering major with an experimental engineering skills course. After successful execution, I promoted a Curriculum Change Proposal to add Engr 201 as a requirement for two engineering majors, currently on track for inclusion in USAFA's Course of Instruction (COI) in Fall 2026. This change will explicitly provide instruction in engineering communication and will better prepare our graduates to lead in technically complex environments.

USAFA has long been a recognized leader in space education. The Astronautical Engineering major was the first of its kind to receive ABET accreditation in 1973. The Department has produced multiple foundational textbooks, as well as a series of successful FalconSAT spacecraft introducing major engineering advances.

While USAFA is well-positioned to maintain its legacy of excellence and leadership in space education, the Department must continue to stay abreast of changes in the field. In the near term, USAFA needs a Curriculum Change Proposal to update the COI's description of the Astronautical Engineering major. The COI contains a list of ABET's program requirements, but ABET has introduced new requirements for 2026 and beyond. Aligning the major to ABET's new requirements will subtly change Astronautics instruction and assessment over time.

If selected as Permanent Professor of Astronautics, I also intend to maintain USAFA's leadership in Space Engineering education with the following initiatives.

- Secure a permanent USSF 13S (Space Operations) billet for the Department to ensure an enduring connection to operational Space Force needs.

- Update textbooks. Many of the field's foundational textbooks were developed by USAFA but are due for a refresh to track developments in a rapidly changing environment.
- Promote regular discourse with the other North American schools offering space engineering programs—continuation of work I began in early 2025.

USAFA provides unparalleled leadership in space education and has taught Astronautical Engineering to more graduates than any other institution. I will be proud to sustain this legacy and ensure USAFA continues to offer the nation's premiere Astronautical Engineering degree.